Team Roles and RACI Matrix

Team: Takım 16 · SENG 321 Software Engineering

1. Team members and roles

Team member	Student no.	Role
Ali Kaan Demiralp	230205034	Lead Software Architect and Technical Lead; DevOps and CI/CD Engineer
Murat Yatağan	220205005	Product Owner
Furkan Berkay Öztürk	210204021	Scrum Master and Agile Process Coach
Duhan Kaan Şenol	220205054	Requirements and UX Engineer
Erol Yağız Gemalmaz	220204057	Quality Assurance and Test Automation Engineer

The project specification defines five core roles for teams of up to six members; the Implementation and Integration Engineer role applies only to the seven-person team. With five members, the combined *Product Owner and Requirements Engineer* role is divided into two roles, so that every member holds a named role. The Lead Architect additionally holds the DevOps role.

2. Role responsibilities

Role	Primary responsibilities	Required outputs
Lead Software Architect and Technical Lead; DevOps and CI/CD Engineer	Architecture strategy, API contract, security design, code review leadership; Docker, GitHub Actions pipeline, deployment	C4 and UML models, OpenAPI contract, CI/CD pipeline, docker-compose deployment
Product Owner	Product vision, backlog ownership, prioritisation, stakeholder validation, acceptance sign-off	Product backlog, priorities, acceptance decisions, scenario content
Scrum Master and Agile Process Coach	Agile process, stand-ups, retrospectives, impediment tracking, board administration, review coordination	Stand-up and retrospective records, RACI upkeep, GitHub Projects board
Requirements and UX Engineer	Requirements elicitation and specification, user stories and acceptance criteria wording, non-functional requirements, UX design	SRS, quality-measure specification, Figma prototypes, user-flow diagrams
Quality Assurance and Test Automation Engineer	Test strategy, test automation, defect tracking, coverage monitoring	Test plan, unit, integration and end-to-end tests, at least 50 % unit-test coverage, SonarQube report

3. Implementation responsibilities

All team members contribute code. Initial assignments:

Team member	Implementation areas
Ali Kaan Demiralp	API skeleton, authentication and authorisation, database, session and scoring logic, Docker, CI/CD
Murat Yatağan	Scenario content (seed library, indicator vocabulary), administrator statistics
Furkan Berkay Öztürk	Administrator scenario-management interface
Duhan Kaan Şenol	Learner interface (scenario view, session summary, score history)
Erol Yağız Gemalmaz	Test suites, test tooling, SonarQube configuration

Each member reviews at least one pull request of another member per week. Assignments may be rebalanced at Sprint Planning.

4. RACI matrix

R = Responsible, A = Accountable (exactly one per activity), C = Consulted, I = Informed.

Columns: **AKD** = Ali Kaan Demiralp (Lead Architect, DevOps), **MY** = Murat Yatağan (Product Owner), **FBÖ** = Furkan Berkay Öztürk (Scrum Master), **DKŞ** = Duhan Kaan Şenol (Requirements and UX), **EYG** = Erol Yağız Gemalmaz (QA).

Activity	AKD	MY	FBÖ	DKŞ	EYG
Project proposal and problem definition	C	**A/R**	C	R	C
License decision record	**A/R**	C	I	I	I
System vision and boundary	**A/R**	C	I	C	I
OpenAPI contract	**A/R**	C	I	C	C
Role assignments and RACI	C	C	**A/R**	I	I
GitHub org, Git-Flow, branch protection	**A/R**	I	C	I	I
Product backlog, user stories, acceptance criteria	C	**A**	C	R	C
Backlog prioritisation, story estimation	C	**A/R**	C	C	C
Sprint planning, stand-up notes, retrospectives	C	C	**A/R**	I	I
Impediment tracking, GitHub Projects board	I	I	**A/R**	I	I
SRS and non-functional requirements (Sprint 2)	C	C	I	**A/R**	C
Figma UI/UX prototypes and user flows (Sprint 2)	C	C	I	**A/R**	C
C4 and UML architecture (Sprint 2)	**A/R**	I	I	C	C
Code review leadership, merge decisions	**A/R**	C	C	C	C
Backend implementation (API, DB, security)	**A/R**	I	I	I	C
Frontend: learner flow	C	I	I	**A/R**	C
Frontend: admin dashboard	C	C	**A/R**	C	C
Scenario content and statistics	C	**A/R**	I	C	C

Activity	AKD	MY	FBÖ	DKŞ	EYG
Test plan, unit, integration, e2e tests	C	I	I	I	**A/R**
Coverage monitoring (≥ 50 %), defect tracking	I	I	I	I	**A/R**
GitHub Actions CI pipeline	**A/R**	I	I	I	C
Docker, docker-compose, deployment	**A/R**	I	I	I	C
SonarQube and tech-debt report	R	I	I	I	**A**
Security basics (hashing, RBAC, validation, rate limiting)	**A/R**	I	I	I	C (tests)
Final demo and architectural defense	**A/R**	R	R	R	R
Sprint submissions (evidence links, team info)	C	R	**A**	R	R

Each activity has exactly one Accountable member. Where the Accountable member is also Responsible, another member reviews the result.

5. Contribution evidence

Individual contribution is evidenced through GitHub issues, commits, pull requests, code reviews, documentation, tests and presentations. Every backlog item has a named owner, every change enters through a reviewed pull request, and the Scrum Master records impediments and process decisions.